

Chemical Engineering Program
CHE 331 Syllabus – Fall 2024

1. Department, Number, and Course Title: CHE 330 Chemical Engineering Lab II
2. Class Time, Days & Location: *Laboratory* 1:30 PM – 5:30 PM, T/ W/ Th/ F
3. Instructors:

Dr. Hassan Golpour

Office: 2086-A EB1, Email: hgolpou@ncsu.edu

Office Hours: Wednesdays & Fridays, 8:45 AM – 10:15 AM

Personal Zoom Link: <https://ncsu.zoom.us/my/hassan.golpour>

4. Laboratory Support Professional:

Dr. Kimberly Dickey

Office: 1016 EB1, Phone: 515-7733, Email: kcdickey@ncsu.edu

Office Hours: Mondays from 11:30 am – 1 pm and Thursdays from 9:30 am – 11 am

Personal Zoom Link: <https://ncsu.zoom.us/my/kimdickey>

5. Teaching Assistants:

Temperature Control: Junbin Li

Email: jli264@ncsu.edu

Office Hours: Wednesdays 9:30 am - 11:00 am and Fridays 10:30 am - 12:00 pm

Zoom Link:

<https://ncsu.zoom.us/j/5879417585?pwd=OHQyQjY2d2w5ZzUyTXFFN2tFcENyZz09>

Meeting ID: 587 941 7585

Passcode: 138477

Batch Distillation: Jose Escarce

Email: jfescarc@ncsu.edu

Office Hours: Tuesday and Thursday from 11:00 am - 12:30 pm

Zoom Link:

<https://ncsu.zoom.us/j/95394676256?pwd=n8rrqVfjL4OB52xeMeo2aE4WEgvsrN.1>

Meeting ID: 953 9467 6256

Passcode: 068872

Air Membrane Separation: John Cheadle

Email: jbcheadl@ncsu.edu

Office Hours: Mondays, Wednesdays, Fridays 11:00 AM - 12:00 PM

Zoom Link:

<https://ncsu.zoom.us/j/7325276829?pwd=bksrV0Q5a2JUOTBZdHdaRElmSzMvUT09>

Meeting ID: 732 527 6829

Passcode: 893790

6. Prerequisite(s): CHE 312, CHE 330

7. Textbook(s) and/or Other Required Material:

- CHE 331 Coursepack (available on the web)
- Michael Alley, The Craft of Scientific Presentations, 2nd edition (2013)
ISBN-10: 1441982787 ISBN-13: 978-1441982780 Cost: \$27.90

8. Course Learning Objectives:

- 1) Produce written technical reports as part of an engineering team.
- 2) Orally present technical information as part of an engineering team.
- 3) Capably perform hands-on engineering as part of an engineering team.
- 4) Identify causes and preventative measures for process safety incidents.

9. Class/Laboratory Schedule (# sessions/week and session duration)

One 4-hour lab period per week. A total of three experiments are spread over the semester. Each experiment is followed by time for report and presentation preparation before a quiz/presentation/preview period.

10. Contribution of the Course to Meeting the Professional Component: Engineering Topics: 2 semester hours

11. Scheduled Laboratory/Assignment Dates:

Note: Laboratory reports are due at 1:30 PM on the date two or three weeks after the experiment is performed, e.g. students in the Wednesday laboratory section must submit laboratory reports on Wednesdays.

Table 1 – CHE 331 Course Schedule

Date	Location and Time	Event / Assignment Due
Aug 27 – 30 (Tue – Fri)	EB1-1016, 1:30 PM	Laboratory #1
Sep 10 – 13 (Tue – Fri)	Moodle, 1:30 PM on Friday TurnItIn.com, 1:30 PM Zoom, 1:30 PM	Quiz #1 Laboratory Report #1 Presentation #1 (in classic CHE 330 format)
Sep 24 – 27 (Tue – Fri)	EB1-1016, 1:30 PM	Laboratory #2
Oct 8 – 11 (Tue – Fri)	Moodle, 1:30 PM on Friday TurnItIn.com, 1:30 PM Zoom, 1:30 PM	Quiz #2 Laboratory Report #2 Presentation #2 (in evidence-assertion format)
Oct 22 – 25 (Tue – Fri)	EB1-1016, 1:30 PM	Laboratory #3
Nov 12 – 15 (Tue – Fri)	Moodle, 1:30 PM on Friday TurnItIn.com, 1:30 PM Zoom, 1:30 PM	Quiz #3 Laboratory Report #3 Poster Presentation #3
Nov 22 (Fri)	TurnItIn.com, 5 PM	Safety Paper

12. Grading

36% 3 Lab Reports (12% each)

36% 2 Presentations & 1 Poster Presentation (12% each)

6% 3 Prelabs (2% each)

15% 3 Experiment Quizzes (5% each)

7% 1 Safety Assignment

An individual's score (IS) on a report is determined based on peer evaluations and the average team score (TS) using the following formula:

$$IS_n = TS(0.50 + 0.50 \frac{(\sum_1^i \frac{IPE_{n,i}}{APE_i^*} - \frac{IPE_{n,n}}{APE_n^*})}{n-1}) \quad (1)$$

where, IS_n = individual score of the n^{th} member, $IPE_{n,i}$ = average peer evaluation score for the n^{th} member by the i^{th} member, $IPE_{n,n}$ = average peer evaluation score for the n^{th} member by themselves, APE_i^* = average peer evaluation score by the i^{th} member excluding the member's evaluation of themselves, and APE_n^* = average peer evaluation score by the n^{th} member by themselves

For example: The first and second group member's score (in a five-person team) would be calculated as follows:

$$IS_1 = TS(0.50 + 0.50(\frac{IPE_{1,2}}{APE_2^*} + \frac{IPE_{1,3}}{APE_3^*} + \frac{IPE_{1,4}}{APE_4^*} + \frac{IPE_{1,5}}{APE_5^*})(\frac{1}{5-1})) \quad (2)$$

$$IS_2 = TS(0.50 + 0.50(\frac{IPE_{2,1}}{APE_1^*} + \frac{IPE_{2,3}}{APE_3^*} + \frac{IPE_{2,4}}{APE_4^*} + \frac{IPE_{2,5}}{APE_5^*})(\frac{1}{5-1})) \quad (3)$$

IPE scores for each team member are awarded by the instructor(s) based on review of the individual's team peer evaluation information.

Course grades will be determined as follows:

Score	>95.5	92.5 to 95.4	89.5 to 92.4	86.5 to 89.4	82.5 to 86.4	79.5 to 82.4	76.5 to 79.4	72.5 to 76.4	69.5 to 72.4	66.5 to 69.4	62.5 to 66.4	59.5 to 62.4	<59.5
Grade	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F

13. Additional Course Information

- Laboratory reports will be submitted electronically by uploading one PDF (preferred format) or MS Word file of the entire laboratory report (e.g. letter of transmittal, report, appendices, laboratory notebook) to the course's TurnItIn.com website by 1:30 PM on the scheduled date of submission. Instances of plagiarism will be penalized on a situational basis and may be referred to the Office of Student Conduct.
- Any delay in report submission will decrease the report grade by 2% (and an additional 2% per each additional hour of delay, up to a maximum of 10% per calendar day). Reports, etc. may be turned in electronically on weekends/holidays to avoid further penalty.
- Preview videos for each experiment in the laboratory are available online via Google

Drive at the following link:

<https://drive.google.com/drive/folders/1bTHPJP-s7zzR0aw3rdHLSerkY7jywtZ6?usp=sharing>

Note, that the above videos have been recorded by previous lab TAs in Fall 2020. Refer to page 1 of this document to find your current TAs for each experiment this semester.

It is the students' responsibility to view these videos on their own outside of class prior to contributing to their group's pre-lab documents and completing each experiment.

- Any student who arrives to any laboratory session more than 10 minutes late without written permission from the instructor or a university-approved excuse will be considered absent from the session and will receive zero credit for deliverables associated with the experiment.
- Students who miss a quiz without prior written permission from Dr. Golpour or a university-approved excuse will receive zero credit for the quiz.
- Students who miss all or a portion of their section's scheduled presentation session without prior written permission from Dr. Golpour or a university-approved excuse will receive zero credit for the presentation.
- Any student who attempts to "steer" selection of their laboratory team members by providing false or misleading information during the CATME team assignment process will be referred to NCSU's Office of Student Conduct.
- Peer evaluation feedback must be submitted via CATME by the time the online survey closes (typically 3-4 days after reports/presentations are submitted). CATME surveys will not be redistributed for a group under any circumstances.
- Dr. Golpour may ask a student group for a "long-form" peer evaluation to clarify CATME scores if (1) one or more team members received a peer evaluation score more than 15% below the group average and (2) little or no written explanation is given in CATME student comments for the low peer evaluation score(s). Student teams can avoid long-form evaluations by explaining peer evaluation scoring in comments through CATME.
- Peer evaluation scores may be adjusted at the discretion of the instructor in circumstances where it appears inappropriate team dynamics have prevented accurate peer evaluations.
- The instructor may assign a grade of zero for a student's report and/or presentation if it is apparent the student did not meaningfully contribute to the group assignment.
- It is prohibited for students in different groups to share graded, ungraded or in-progress course materials (including, but not limited to, reports, notes from Q&A sessions, homework, etc.) in any format. Giving another student(s) any course materials without instructor consent is considered "Aiding and Abetting" cheating as per NCSU's Code of Student Conduct. In other words, if a student in another group is found to have plagiarized your report, used your notes from Q&A sessions, etc., your grades will be eligible to be reduced by the Office of Student Conduct (OSC). If this occurs after you complete CHE 330, your grade is eligible to be retroactively reduced (possibly to an F, which would

void your CBE degree if you've already graduated).

- All electronic and hard-copy documents (including but not limited to: copies of lecture notes, homework assignments and solutions, exam assignments and solutions, graded/ungraded reports, handouts provided by the instructor, etc.) made available to students in this course are copyrighted and intended only for students' personal use. It is prohibited to share any content from the class with a third party (e.g. any person not signed up for the course this semester, personal / public / commercial websites, news or advertising media, etc.) without written permission from the course instructor. Students who distribute course materials to third parties without appropriate permission will receive a grade of incomplete (IN) for the course.
- It is prohibited to make audio or video recordings of the course instructor without prior written approval.
- Students who decide to take this course as "pass/fail" at any point during the semester will not be permitted to participate in the laboratory portion of the course.
- As much as possible graded reports will be returned within two weeks (exclusive of holidays / university closures) after they have been submitted.
- Office hours will not be held on dates when classes are not in session.
- If you believe that an error was made in grading a quiz, write a short justification of your claim and attach it to the original assignment in question. Put the justification (with your contact email address) and assignment in Dr. Golpour's faculty mailbox. The TA or instructor will review your concern and respond to you directly. The "statute of limitations" for submitting such claims is one week after the assignment is returned.
- Grading laboratory reports involves both objective and subjective judgments on the part of the grader. If you believe that an error in objective judgment (AKA an error of fact) was made in grading a report, write a short justification of your claim and put the justification (with your contact email address) in Dr. Golpour's faculty mailbox. The instructor will review your concern and respond to you directly. The "statute of limitations" for submitting such claims is one week after the report is returned. Subjective judgments in report grading are not eligible for regrade consideration. The following are examples of objective and subjective judgments by the grader:
 - Examples of objective judgments (eligible for regrade):
 - Points were deducted for a missing laboratory notebook when the notebook was in fact included in the report appendix.
 - Points were deducted for not discussing a specific phenomenon when the discussion was in fact included in the report, but overlooked by the grader.
 - Examples of subjective judgments (ineligible for regrade):
 - Points were deducted for lack of clarity in writing.
 - Points were deducted for lack of effectiveness of a figure or table.
- Cell phones, laptops, tablet computers, etc. must be turned off and put away

during quizzes.

- All students registered with DSO should make arrangements to take tests at the DSO testing center and the instructor should be notified in advance of any arrangements for testing. Students who are registered with DSO but do not make reservations with the DSO office more than three business days in advance of the test date can still inquire about the possibility of receiving their accommodations, but should plan to test with the rest of the class and without accommodations. DSO space (and faculty availability) is often minimal ahead of the three-day online scheduling cut-off so students should make all of their reservations as early in the semester as possible.
- Bonus points may be awarded by the instructor(s) for extraordinary achievement of a student or a team.
- There will be no final examination.

14. Use of AI:

No tool can replace your own thinking or analysis. (This includes generative AI such as Chat GPT and “tutoring” sites like Chegg.) Scientific calculators and the mathematical software taught in the class are sufficient for the assignments in this class. Students should check with their instructor before using other technologies. Artificial intelligence (AI) language tools, such as (but not limited to) ChatGPT, Dall-e, etc. may not be used for course assignments except as explicitly authorized by the instructor.

- To cite any generative AI, you must include the prompt you entered. For example, “Describe the symbolism of the green light in the book *The Great Gatsby* by F. Scott Fitzgerald” prompt. ChatGPT, Feb 13, 2024. version, OpenAI, as released 8 Mar. 2023, chat.openai.com/chat.” Any assignment that is found to have used generative AI tools in unauthorized ways is a violation of academic misconduct. When in doubt about permitted usage, please ask the instructor for clarification.
- AI is not designed for and should not be used to solve engineering problems as a stand-alone product. If you use any generative AI in your work, provide evidence of
 - how you used the AI tool,
 - how you verified its accuracy and reliability,
 - how you integrated its output with your own work, and
 - how you acknowledged its limitations and biases.

The use of generative AI tools (e.g. ChatGPT, Dall-e, etc.) is permitted in this course for the following activities:

- Brainstorming and refining your ideas
- Fine-tuning your research questions
- Finding information on your topic
- Drafting an outline to organize your thoughts
- Checking grammar and style

The use of generative AI tools is not permitted in this course for the following activities:

- Writing a draft of a writing assignment
- Writing entire sentences, paragraphs, or papers to complete class assignments
- Paraphrasing a sentence or paragraph
- Solving homework and quiz problems

You are responsible for the information you submit based on an AI query (for instance, that it does not violate intellectual property laws, or contain misinformation or unethical content). Your use of AI tools must be properly documented and cited in order to stay within university policies on academic honesty.

15. Additional Required Text for NCSU Syllabi

- **Academic Integrity:** Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct found at <http://policies.ncsu.edu/policy/pol-11-35-01>. It is the instructor's understanding and expectation that the student's signature on any test or assignment means that the student contributed to the assignment in question (if a group assignment) and that they neither gave nor received unauthorized aid. Authorized aid on an individual assignment includes discussing the interpretation of the problem statement, sharing ideas or approaches for solving the problem, and explaining concepts involved in the problem. Any other aid would be unauthorized and a violation of the academic integrity policy. Unauthorized aid additionally includes accessing on-line solutions, whether that be complete or partial on-line copies of textbook problem solutions or paying someone to complete your homework. In addition, *any computer work submitted must be completed on your own personal computer or from your own @ncsu.edu account to avoid confusion about the origin of the file, and no sharing of files in any way is allowed.* All cases of academic misconduct will be submitted to the Office of Student Conduct. Students found guilty of academic misconduct will be subject to, at a minimum, a zero on the assignment in question, up to a zero for that course component (e.g. a zero for the homework portion of the final grade), or a failing grade in the course, depending on the nature of the violation. In addition, if you are found guilty of academic misconduct in the course, you will be on academic integrity probation for the remainder of your years at NCSU and may be required to report your violation on future professional school applications.
- **Students With Disabilities:** Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Services Office at Suite 2221, Student Health Center, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the **Academic Accommodations for Students with Disabilities Regulation (REG02.20.01)**(<http://policies.ncsu.edu/regulation/reg-02-20-01>)
- **Non-Self-Contained Laboratory:** The standard time associated with a 2-hour non-self-contained laboratory is 3.33 hours. Depending on the pace at which students complete experiments, it is possible completion of some laboratory procedures will

take up to 4 hours, but historically most laboratories are completed within 3.33 hours. Students will not report to the laboratory during weeks experiments are not scheduled and are encouraged to use these time periods to work on reports and other deliverables associated with the laboratory.

- **Attendance Regulation:** <http://policies.ncsu.edu/regulation/reg-02-20-03>
- **Equal Opportunity:** NC State University provides equality of opportunity in education and employment for all students and employees. Accordingly, NC State affirms its commitment to maintain a work environment for all employees and an academic environment for all students that is free from all forms of discrimination. Discrimination based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law and/or NC State University policy and will not be tolerated. Harassment of any person (either in the form of quid pro quo or creation of a hostile environment) based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation also is a violation of state and federal law and/or NC State University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at <http://policies.ncsu.edu/policy/pol-04-25-05> or http://www.ncsu.edu/equal_op/. Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148.
- **NC State Policies, Regulations, and Rules (PRR):** Students are responsible for reviewing the NC State University PRR's located at <http://oucc.ncsu.edu/course-rights-and-responsibilities> which pertains to their course rights and responsibilities.
- **ClassEval:** Course and instructor evaluations: Online class evaluations will be available for students to complete during the last two weeks of class. Students will receive an email message directing them to a website where they can login using their Unity ID and complete evaluations. All evaluations are confidential; instructors will never know how any one student responded to any question, and students will never know the ratings for any particular instructors.
 - Evaluation website: <https://classeval.ncsu.edu>
 - More information: <http://www2.acs.ncsu.edu/UPA/classeval/index.htm>
 - Schedule: Online class evaluations will be available for students to complete during the last 2 weeks of class and become unavailable before finals begin.

16. COVID-19 Policy

Due to the Coronavirus pandemic, public health measures have been implemented across campus. Students should stay current with these practices and expectations through the [Protect the Pack](https://www.ncsu.edu/coronavirus/) website (<https://www.ncsu.edu/coronavirus/>). The sections below provide expectations and conduct related to COVID-19 issues

Health and Participation in Class

We are most concerned about your health and the health of your classmates and instructors/TAs.

- If you test positive for COVID-19, or are told by a healthcare provider that you are presumed positive for the virus, please work with your instructor on health accommodations and follow other university guidelines, including self-reporting: <https://healthypack.dasa.ncsu.edu/coronavirus/>. Self-reporting is not only to help provide support to you, but also to assist in contact tracing for containing the spread of the virus.
- If you feel unwell, even if you have not been knowingly exposed to COVID-19, please do not come to class.
- If you are in quarantine, have been notified that you may have been exposed to COVID-19, or have a personal or family situation related to COVID-19 that prevents you from attending this course in person (or synchronously), please connect with your instructor to discuss the situation and make alternative plans, as necessary.
- If you need to make a request for an academic consideration related to COVID-19, such as a discussion about possible options for remote learning, please talk with your instructor for the appropriate process to make a COVID-19 request.

Health and Well-Being Resources

These are difficult times, and academic and personal stress is a natural result. Everyone is encouraged to take care of themselves and their peers. If you need additional support, there are many resources on campus to help you:

- Counseling Center (<https://counseling.dasa.ncsu.edu/>)
- Health Center (<https://healthypack.dasa.ncsu.edu/>)
- If the personal behavior of a classmate concerns or worries you, either for the classmate's well-being or yours, we encourage you to report this behavior to the NC State CARES team: (<https://advising.dasa.ncsu.edu/resources-for-advisors/advisors-toolkit/cares/>)
- If you or someone you know are experiencing food, housing or financial insecurity, please see the Pack Essentials Program (<https://dasa.ncsu.edu/pack-essentials/>).

Community Standards related to COVID-19

We are all responsible for protecting ourselves and our community. Please see the community expectations and Rule 04.21.01 regarding Personal Safety Requirements Related to COVID-19 <https://policies.ncsu.edu/rule/rul-04-21-01/>

Course Expectations Related to COVID-19:

- Personal Protective Equipment: As a member of the NC State academic community you are required to follow all university guidelines for personal safety with face coverings, physical distancing, and sanitation. Face coverings are required in this class and in all NC State buildings. Face coverings should be worn to cover the nose and mouth and be close fitting to the face with minimal gaps on the sides. In addition, students are responsible for keeping their course/work area clean.
- Course Attendance: NC State attendance policies can be found at: <https://policies.ncsu.edu/regulation/reg-02-20-03-attendance-regulations/>. Please refer to this course's attendance, absence, and deadline policies for additional details. If you are quarantined or otherwise need to miss class because you have been advised that you may have been exposed to COVID-19, you should not be penalized regarding attendance or class participation. However, you will be expected to develop a plan to keep up with your coursework during any such absences. If you become ill with COVID-19, you should follow the steps outlined in the health and participating section above. COVID 19-related absences will be considered excused; documentation need only involve communication with your instructor.
- Course Meeting Schedule: Your course might not have a traditional meeting schedule in Fall 2020. Be sure to pay attention to any updates to the course schedule as the information in this syllabus may have changed. Please discuss any questions you have with the instructor.

- **Classroom Seating:** To support efficient, effective contact tracing, please sit in the same seat when possible and take note of who is sitting around you; instructors may also assign seats for this purpose..
- **Technology Requirements:** This course may require particular technologies to complete coursework. Be sure to review the syllabus for these expectations, and see go.ncsu.edu/syllabus-tech-requirements to find out more about technical requirements for your course. If you need access to additional technological support, please contact the Libraries' Technology Lending Service: <https://www.lib.ncsu.edu/devices>.

Course Delivery Changes Related to COVID-19

Please be aware that the situation regarding COVID-19 is frequently changing, and the delivery mode of this course may need to change accordingly, including from in-person to online. Regardless of the delivery method, we will strive to provide a high-quality learning experience.

Grading/Scheduling Changing Options Related to COVID-19

If the delivery mode has a negative impact on your academic performance in this course, the university has provided tools to potentially reduce the impact:

- **Enhanced S/U Grading Option:** <https://studentservices.ncsu.edu/your-resources/covid-19/spring2020-sat-grading/>
- **Late Drop:** <https://studentservices.ncsu.edu/your-resources/covid-19/spring2020-latedrop/>

In some cases, another option may be to request an incomplete in the course. Before using any of these tools, discuss the options with your instructor and your academic advisor. Be aware that if you use the enhanced S/U, you will still need to complete the course and receive at least a C- to pass the course.

Other Important Resources

- Keep Learning: <https://dasa.ncsu.edu/students/keep-learning/>
- Protect the Pack FAQs: <https://www.ncsu.edu/coronavirus/frequently-asked-questions/>
- NC State Protect the Pack Resources for Students: <https://www.ncsu.edu/coronavirus/reactivating-campus/resources-for-students/>
- NC State Keep Learning, tips for students opting to take courses remotely: <https://dasa.ncsu.edu/students/keep-learning/>
- Introduction to Zoom for students: <https://youtu.be/5LbPzzPbYEW>
- Learning with Moodle, a student's guide to using Moodle: <https://moodle-projects.wolfware.ncsu.edu/course/view.php?id=226>